

16.00 Benoît Timmermans (*Université Libre de Bruxelles*): Espace et représentation: la tension philosophique entre géométrie et théorie des groupes chez Weyl

16.45 Philippe Nabonnand (*Archives Poincaré, Nancy*): Les connections à partir de la correspondance entre Elie Cartan et Hermann Weyl

19.30 Special Evening. Visit of the Art-and-Philosophy Exhibition: *Gedankengeflechte. Ein philosophischer Blick durch abstrakte und organische Kunst.* (BildungsTURM, Konstanz)*

20.30. Dinner at the restaurant *Hafenhalle**

Friday 29 May 2015

Morning session:

The Changing Faces of Subjectivity in Weyl's Philosophy

Chair: E.-M. Engelen (*University of Konstanz*)

9.30 Norman Sieroka (*ETH, Zürich*): Hermann Weyl's Neighbourhoods: 'Spaces' in Mathematics, Physics, Subjectivity, and Historiography

10.15 Dominique Pradelle (*Université de la Sorbonne-Paris IV*): Le problème de la définition du continu : entre phénoménologie et intuitionnisme

11.00 Tea/coffee break

11.15 Francesca Biagioli (*Zukunftskolleg, Konstanz*): Intuition and Conceptual Construction in Weyl's Analysis of the Problem of Space

12.00 Jairo da Silva (*University of São Paulo*): Husserl and Weyl on the Constitution of Space

12.45 Meal at Mensa (Room K7, self-service)

There is no afternoon session.
Go, visit and enjoy Konstanz!

Practical information: The workshop takes place at the University of Constance, Room V1001, Universitätsstraße 10, in Konstanz.

Bus lines: 9a et 9b

For more practical information please visit our website : www.weylworkshop.com

(*) For further request, especially reservations at the meals (buffet, lunch, dinner) and visit at the BildungsTURM, please contact Christoph Staufenbiel: christoph.staufenbiel@uni-konstanz.de

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Acknowledgments:

The organising institutions are the *University of Constance* (*Zukunftskolleg, Mentorship Program, Philosophy Department*), the CIPh (*Collège International de Philosophie, Paris*), the CEPERC (*Centre d'EPistémologie et d'ERGologie Comparatives, Aix-en-Provence*).

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INTERNATIONAL WORKSHOP

WEYL AND THE PROBLEM OF SPACE

FROM SCIENCE TO PHILOSOPHY

27, 28 & 29 May 2015



University of Constance
(Room V1001)
Universitätsstraße 10 ,
78464 Konstanz, Germany

WEYL AND THE PROBLEM OF SPACE FROM SCIENCE TO PHILOSOPHY

The problem of space is at the core of Hermann Weyl's philosophy. During the years 1916-1923, his scientific work was focused on the foundation of differential geometry and general relativity. In particular, Weyl reformulated the mathematical problem of space in connection to the new relativistic context and solved it. With a Kantian starting point, mediated by Helmholtz, the philosophical problem of space is reformulated by Weyl in the following terms: among all the infinitely many conceivable metrical spaces, which one applies to the physical world? In agreement with Riemann's prophecy, Weyl acknowledges that the metric can quantitatively vary with the physical situation. Despite this, Weyl "deduces", through the analysis of the group of rotations, that there is only one *kind* of metric. But the question remains: how can we correlate this solution with the other aspects of the concept of space (psychophysical and/or phenomenological)? The reasons that underpins this "deduction" leads from philosophy (Kant, Leibniz, Husserl, Fichte, etc.) to physics (Galileo, Einstein, etc.) and conversely, always through the mediation of mathematics.

In a different scientific context, the understanding of this exemplary double move is still of major interest, not only from an historical point of view, but for those who think that without being reducible to one another, the exchanges between science and philosophy must be continued.

Scientific organisation :

Julien Bernard: *Zukunftskolleg*, Konstanz and *CEPERC*, Aix-en-Provence. Carlos Lobo: *Collège International de Philosophie*, Paris and *CFCUL*, Lisbon.

This workshop is linked to the research program : "La question de l'espace comme carrefour épistémologique"
http://www.ciph.org/Site/Carlos_LOBO.html

Wednesday
27 May 2015
(Room, V1001)

13.00 Registration & Welcome (Room, V1002)

13.30. Opening: G. Galizia, D. Emundts, C. Lobo, J. Bernard

Afternoon Session:

The Problem of Space from a Physical Point of View

Chair: **Dina Emundts** (*University of Konstanz*)

14.00 Erhard Scholz (*University of Wuppertal*): Context Dependence and Invariant Aspects in Weyl's View of the Space Problem

14.45 Alexander Afriat (*SPHERE, University of Brest*): Logic of Gauge

15.30 Tea/coffee break

15.45 Charles Alunni (*E.N.S., Paris, Labex-TransferS*): Hermann Weyl chez Gaston Bachelard. Un héritage riemannien

16.30 Silvia de Bianchi (*Universitat Autònoma de Barcelona*): From the Problem of Space to the Epistemology of Science

17.15 Tea/coffee break

17.30 Thomas Ryckman (*Stanford University*): A Weylian Perspective on the Infinite Space of Eternal Inflation

18.30 Buffet* (Room V1002)

* For the buffet on the 27th, lunches and/or dinner on the 28th please contact us for reservation. See below *Practical information*.

Thursday
28 May 2015
(Room, V1001)

Morning Session:

Roundtable on the French-German Edition of *Mathematische Analyse des Raumproblems (MARP)*

Chair: **Andrea Lailach-Hennrich** (*University of Konstanz*)

9.00 Julien Bernard (*Zukunftskolleg and CEPERC*): Historical and Philosophical Considerations on Weyl's *MARP*

9.45 Carlos Lobo (*CIPh, Paris & CFCUL, Lisbonne*): The Philosophical Residuum of the Problem of Space after Weyl's *MARP*

10.30 Eric Audureau (*CEPERC, Aix-en-Provence*): Some Difficulties in Weyl's Solution to the Philosophical Problem of Space

11.15 Tea/coffee break

11.30 General discussion

11h15 Meal at Mensa* (Room K7, special menu)

Afternoon session:

The Problem of Space from a Mathematical Point of View

Chair: **Erhard Scholz** (*University of Wuppertal*)

14.00 Christophe Eckes (*Archives Poincaré, Nancy*): Groups and Manifolds: The Variety of Space Problems in Weyl's work (1913-1927)

14.45 Mark van Atten (*SND, Paris*): Can I Count Myself a King of Infinitesimal Space?

15.30 Coffee-break